Nexus² / Arena

Frontend Product Walkthrough & Architecture Report

March 20, 2026

Current State: Frontend Prototype (No Backend)

1. Executive Summary

Nexus² (also referred to as Arena) is a privacy-first learning companion application designed for university and graduate-level students. It provides a structured environment for engaging with course materials through an AI tutor named Nexi, organized note-taking, personal reflections, study task management, and progress tracking.

The product is built as a single-page React application using Vite, TypeScript, Tailwind CSS, and shadcn/ui components. All data is currently stored in-memory via React context (WorkspaceContext) with no backend or database. The UI is fully functional as an interactive prototype with simulated AI responses and seed data.

The design direction is intentionally warm, calm, and scholarly. It avoids gamification, neon colors, and generic AI aesthetics. Light mode uses a warm paper-like palette; dark mode uses warm charcoal tones. Typography combines Source Serif 4 for headings with Inter for body text.

Key metrics at a glance

- 9 routes: Home, Library, Workspace, Notebook, Reflections, Progress, Study Plan, Settings, Profile
- 3-pane workspace layout with Sources, Nexi AI chat, and Notebook sidebar
- 5-tab Settings page redesigned today with visual card pickers
- Theme system supports Light, Dark, and Auto (system) modes
- 4 font family options: Default, Sans, Serif, Dyslexic-friendly
- 10-language selector in sidebar user menu
- All state is in-memory (no persistence across page reloads)

2. Product Identity

2.1 Core Purpose

Nexus² is a course-centered learning workspace. Unlike generic note-taking apps or chatbots, it structures the learning experience around specific courses and their source materials. The AI tutor (Nexi) is grounded in uploaded course materials, not general internet knowledge. The notebook captures insights during study. Reflections provide a private journal for metacognition.

2.2 Design Philosophy

- Privacy-first: All data stays local. No tracking, no analytics, no third-party sharing. Privacy badges and disclaimers appear throughout the UI.
- No gamification: No points, streaks, badges, or leaderboards. Progress is shown as genuine academic progress, not game mechanics.
- Warm and scholarly: The visual identity draws from paper, libraries, and calm study environments. Nothing feels like a tech dashboard.
- Course-centered: Everything revolves around courses. Sources belong to courses. Nexi conversations are per-course. Notes link back to courses.
- Notebook-style: The product feels like a digital study notebook with an AI reading companion, not a chatbot with a sidebar.

2.3 Visual Identity

The product uses a warm, muted color palette. In light mode, the background is a subtle warm paper tone (HSL 40 20% 98%). The primary color is a muted navy (HSL 220 40% 25%). The accent is a warm amber (HSL 35 60% 52%). In dark mode, the background shifts to warm charcoal (HSL 225 20% 10%) with a warm amber primary.

All colors are defined as HSL CSS custom properties in index.css and referenced through Tailwind's semantic token system. No colors are hardcoded in components.

3. Global Design and App Shell

3.1 Layout Architecture

The app shell is defined in Layout.tsx. It wraps all routes (except 404) in a WorkspaceProvider (global state) and a SidebarProvider (sidebar collapse state). The layout is a horizontal flex container: a fixed-width sidebar on the left and a flexible main content area on the right.

The Layout component also reads the current font family and font size settings from WorkspaceContext, applying a CSS class (font-family-sans, font-family-serif, or font-family-dyslexic) and a CSS variable (--font-scale) to the root container. The main content area uses a font-scaled class that applies zoom-based scaling.

3.2 WorkspaceProvider

WorkspaceContext.tsx is the central state store. It holds:

- Notebook entries (array of notes with title, snippet, course, tags, source, date)
- Chat messages (keyed by course ID, each message has role, content, citations)
- Active source selection
- Reflections (text, mood, linked course, date)
- User profile (name, email, bio, institution, timezone, learning goal)
- App settings (compact mode, font size, font family, language)
- Study tasks (title, course, priority, completed, due date)

All data is seeded with realistic examples (machine learning, statistics, philosophy courses). There is no persistence; refreshing the page resets to seed data.

3.3 Routing

React Router v6 handles navigation. All main routes are nested under the Layout component, which provides the sidebar. The Workspace route uses a dynamic :id parameter. A catch-all 404 route exists outside the layout.

4. Navigation and Sidebar

4.1 Sidebar Structure

The sidebar (AppSidebar.tsx) uses shadcn's collapsible sidebar system with icon-rail mode. It has three sections:

Header: Displays the Nexus² logo text (or N² when collapsed) and a collapse/expand toggle button using chevron icons.

Main Navigation: Six items in the following order (reordered during today's session): Home, Library, Study Plan, Notebook, Progress, Reflections.

Footer: A theme toggle button (sun/moon icon) and the user menu (SidebarUserMenu).

4.2 Expanded vs Collapsed Behavior

When expanded, the sidebar shows icon + label for each nav item, the full Nexus² text logo, and user name/plan in the footer. When collapsed to icon-rail mode, only icons are visible with tooltips on hover. The user avatar replaces the name/email. The collapse toggle changes from left-pointing chevrons to right-pointing chevrons.

The sidebar supports keyboard shortcut Ctrl/Cmd+B to toggle collapse state. State is persisted via a cookie (sidebar:state).

4.3 Active States

The active route is highlighted with a subtle background tint (sidebar-accent) and medium font weight. The isActive function matches exact path for Home (/) and startsWith for other routes.

4.4 User Menu

SidebarUserMenu renders a dropdown menu triggered by clicking the user's avatar/name area. The dropdown contains:

- Header with avatar, name, and email
- Profile link (navigates to /profile)
- Settings link (navigates to /settings)
- Language submenu with 10 languages (English, Español, Français, Deutsch, Português, Japanese, Chinese, Korean, Arabic, Hindi). Selected language shows a checkmark. Selecting a language updates appSettings.language and shows a toast confirmation.
- Get Help (shows coming soon toast)
- Learn More submenu (About, Tutorials, Keyboard shortcuts, Usage/Privacy policy)
- Log Out (shows farewell toast, no actual auth)

Important: The sidebar language setting controls the UI language preference, stored separately from Nexi's preferred explanation language in Settings. They are intentionally independent.

5. Theme and Typography

5.1 Theme Modes

ThemeProvider.tsx manages three theme preferences: Light, Dark, and Auto. The “Auto” mode was added during today’s session.

- **Light:** Warm paper background, muted navy text, amber accents. Cards have subtle warm shadows. Everything feels like high-quality stationery.
- **Dark:** Warm charcoal background (not pure black). Text shifts to warm off-white. Amber accents remain. Cards use darker borders. The overall feel stays warm and readable.
- **Auto:** Follows the operating system’s color scheme preference (prefers-color-scheme media query). A real-time listener updates the theme immediately if the OS preference changes while the app is open.

Theme preference is stored in localStorage under the key nexus-theme. The resolved theme (always light or dark) is applied as a CSS class on the HTML element, which triggers Tailwind’s dark mode variants.

5.2 Typography

The app uses a deliberate typographic hierarchy:

- **Source Serif 4:** Used for page headings (h1 elements) via the font-serif Tailwind class. Gives headings a scholarly, editorial quality.
- **Inter:** Used for all body text, labels, buttons, metadata, and UI controls via font-sans. Clean and highly readable at small sizes.
- **OpenDyslexic:** Available as an accessibility option. Loaded via Google Fonts. Applied globally when selected in Settings.

Font family selection was added today. Users can choose from four options in Settings: Default (Source Serif 4 for headings, Inter for body), Sans (Inter everywhere), Serif (Source Serif 4 everywhere), and Dyslexic (OpenDyslexic everywhere). The selection applies a CSS class to the root layout container.

5.3 Font Size

Three sizes are available: Small (0.9x zoom), Medium (1x, default), and Large (1.12x zoom). This is implemented via a CSS variable --font-scale applied through the zoom CSS property on the main content area.

6. Page-by-Page Walkthrough

6.1 Home Page

The Home page (/pages/Index.tsx) serves as a calm re-entry surface. It is not a dashboard. It greets the user and helps them resume their work.

What the user sees

- A time-based greeting: Good morning, Good afternoon, or Good evening, followed by the user's first name.
- A subtitle: Welcome back to your workspace.
- A prominent Continue Learning card showing the most recent course (title, current module, progress percentage, and a progress bar). Clicking it navigates to the workspace.
- An Upcoming Tasks section showing up to 4 incomplete tasks with priority dots (colored circles), titles, and clickable checkboxes to complete them inline.
- A Recent Workspaces section with smaller course cards showing last-active time and a progress bar.
- Quick Actions: three buttons linking to Library, Notebook, and Study Plan.

The data is partially hardcoded (course list) and partially dynamic (tasks come from WorkspaceContext, greeting is time-based, user name comes from profile).

6.2 Library Page

The Library page (/pages/Library.tsx) is the course directory. It lists all courses the user has access to.

What the user sees

- A serif heading (Library) with a subtitle.
- A search input to filter courses by title.
- Four filter tabs: All, Active, Completed, Pinned.
- A responsive grid of course cards. Each card shows: title, description, progress bar with percentage, last-accessed time, and a Complete badge if finished. Pinned courses show a pin icon.
- Clicking a course card navigates to /workspace/:id.

Course data is hardcoded (3 courses). The search and filter controls are fully functional against the static data.

6.3 Study Plan Page

The Study Plan page (/pages/StudyPlan.tsx) is a task management interface for academic work.

What the user sees

- A serif heading with a dynamic count of remaining tasks.
- Two header buttons: Sync (toggles a sync panel for Google Calendar and Notion integration) and Add Task (opens an inline form).

- The Add Task form includes: text input, priority selector (High/Medium/Low), course selector, date picker (calendar popover), and time picker (appears after selecting a date). Enter key or Add button submits.
- Active tasks listed by priority (high first). Each row shows a circular checkbox, title, priority dot with label, optional course badge, optional due date, and a delete button that appears on hover.
- Completed tasks are hidden behind a collapsible toggle showing the count.
- An empty state appears when all tasks are done: a checkmark icon and congratulatory text.

Task state (add, complete, delete) is fully dynamic through WorkspaceContext. Sync toggles are UI-only (no actual integration). The date picker uses the date-fns library and shadcn's Calendar component.

6.4 Notebook Page

The Notebook page (/pages/Notebook.tsx) is the full-page note management experience, separate from the workspace sidebar pane.

What the user sees

- Search bar to filter notes by title or content.
- Sort options: Recent, by Course, by Tag.
- View mode toggle between list and card layout.
- Notes displayed with title, snippet, course tag (color-coded), additional tags, and date.
- Clicking a note opens a full-screen editor overlay with: a formatting toolbar (bold, italic, underline, lists), a title field, a course selector dropdown, tag editing, and a large textarea for content.
- A New Note button opens a blank editor. Save writes to WorkspaceContext. Delete removes the entry.

All operations (create, edit, delete, search, sort) are fully functional. Notes persist only within the current browser session.

6.5 Progress Page

The Progress page (/pages/Progress.tsx) shows learning analytics and course completion data.

What the user sees

- A hero summary card showing: current streak (days), active courses count, total study hours, and courses needing attention.
- A weekly activity bar chart showing hours studied per day (Mon–Sun).
- Individual course progress cards with: title, module info, progress bar (animated on scroll), and a warning icon for courses needing attention.
- Scroll reveal animations using the useScrollReveal hook (Intersection Observer-based).

All data on this page is hardcoded. Progress bars animate when they enter the viewport. The page functions as a visual prototype showing what the analytics view would look like with real data.

Known weakness: The hero summary metrics use round numbers. The weekly activity chart is a simple bar chart made with plain divs, not a charting library. This page would benefit from a redesign with richer visualizations when real data is available.

6.6 Reflections Page

The Reflections page (/pages/Reflections.tsx) is a private journaling space for learning metacognition.

What the user sees

- A text area at the top for writing new reflections, with Enter to save and Shift+Enter for new lines.
- Prompt chips below the text area (pre-written prompts like What clicked today? that auto-fill the textarea).
- Mood selector: 6 options (Focused, Confused, Energized, Drained, Curious, Calm), each shown as an emoji pill.
- A microphone button for speech-to-text input (uses the browser's SpeechRecognition API where available).
- Saved reflections grouped by date (Today, Yesterday, Earlier) in a vertical timeline.
- Each reflection card shows: content text, optional mood emoji, optional linked course, date, and a delete button (appears on hover).
- Empty state with illustration when no reflections exist.

All operations are dynamic through WorkspaceContext. Speech recognition works in supported browsers (Chrome). Mood tags and course links are optional.

6.7 Settings Page

The Settings page (/pages/Settings.tsx) was significantly redesigned during today's session. It uses a 5-tab layout with a left sidebar navigation.

Tab structure

- **General:** Appearance and workspace preferences (redesigned today with visual card pickers).
- **Nexi:** AI companion behavior configuration.
- **Account:** Organization details and session management.
- **Privacy:** Security and data controls.
- **Connectors:** Third-party integration management.

General tab (redesigned today)

The General tab now features visual card-based pickers instead of simple toggles:

- **Color Mode:** Three selectable cards (Light, Auto, Dark), each roughly 120px wide with rounded-xl borders. Each card contains a mini-illustration showing what that theme looks like (warm paper for Light, split for Auto, charcoal for Dark). The selected card gets an accent-colored border and a small checkmark badge in the top-right corner. Cards have hover effects (slight scale) and active press feedback (scale 0.97).
- **Font Style:** Four Aa preview cards (Default, Sans, Serif, Dyslexic). Each card renders the letters Aa in its actual font family so users can preview the typeface. Selection behavior matches the theme cards.
- **Display section:** Font size S/M/L segmented control and a compact mode toggle, grouped in a bordered card.

- **Voice & Input:** Toggle for voice input (microphone).
- **Reading & Sources:** Toggles for auto-expand sources and auto-save notes.

Nexi tab

Configures the AI companion's behavior:

- Response style: Concise / Detailed / Socratic segmented control.
- Show citations toggle.
- Follow-up suggestions toggle.
- Conversation memory toggle.
- Code explanation depth: Beginner / Intermediate / Advanced segmented control.
- Preferred language for Nexi responses: dropdown with 10 languages. This is independent from the sidebar UI language preference.

Account tab

Shows organization ID (copyable), organization name, and a table of active sessions with device, location, creation date, and last activity. Includes a Log Out button.

Privacy tab

Toggles for two-factor authentication, local-only mode, and data sharing. Export data and Reset all buttons with confirmation toasts.

Connectors tab

Lists 6 integration options: Notion, Zotero, Readwise, Obsidian, Google Drive, Google Calendar. Each shows an SVG logo, name, description, and a Connect button. These are UI-only; no actual integrations are wired.

6.8 Profile Page

The Profile page (/pages/Profile.tsx) is a standalone page separate from Settings. It shows the user's learning identity.

What the user sees

- Avatar with initials, name, and a Private / Local only badge.
- View mode shows: email, institution, timezone, bio, and learning goal as detail rows.
- Edit mode: all fields become editable inputs (multiline for bio and goal). Save/Cancel buttons.
- Stats section: 3 courses, a placeholder study time, note count (from WorkspaceContext), and reflection count.
- Privacy notice at the bottom reinforcing that data stays local.

Profile data is stored in WorkspaceContext and fully editable within the session.

7. Core Workspace Walkthrough

The Workspace (/pages/Workspace.tsx) is the core learning environment. It opens when a user clicks a course from Home or Library. It uses a full-screen 3-pane layout without the global sidebar (the workspace takes the entire viewport width).

7.1 Layout Mechanics

Three panes sit side by side in a horizontal flex container:

- **Sources (left):** Variable width. Three modes: mini (56px icon rail), list (260px), and viewer (42% with min/max constraints).
- **Nexi (center):** Flex-grow, always takes remaining space. This is the dominant pane.
- **Notebook (right):** Two states: mini (56px icon rail) or expanded (300px).

Width transitions use the pane-transition CSS class (500ms ease-spring). When the source viewer opens, the notebook automatically collapses to mini to make room. When the viewer closes, the notebook restores to its previous state.

7.2 Sources Pane

Mini mode

Shows a vertical icon rail: a toggle button and a folder icon with a tooltip showing the source count. Clicking the toggle or icon expands to list mode.

List mode (260px)

Displays the course title at top, then source items grouped by module. Each source shows:

- A type-specific icon (video, PDF, slides, code, etc. — 10 source types supported)
- Title and module name
- A colored left-border indicator based on type

Clicking a source item opens the viewer mode.

Viewer mode (42% width)

Shows a back button, the source title and metadata (type badge, read time estimate, module name). Below is the source content area:

- For video sources: a placeholder video player area.
- For link sources: a placeholder external link frame.
- For slides: a placeholder slide viewer.
- For PDF/reading/lecture types: rendered text content with headings, paragraphs, and code blocks.
- Code blocks are parsed and displayed with a filename header and monospace font.

When the viewer opens, the notebook pane automatically collapses to mini mode to provide more horizontal space.

Source data is hardcoded (two modules with 5 sources each for the ML course). Source document content is also hardcoded with realistic academic text.

7.3 Nexi Pane (Center)

Nexi is the AI tutor at the heart of the workspace. The center pane provides a chat-style interface.

Visual design

The pane has a header showing the course title, current module, and a Nexi badge with a sparkle icon. The chat area is a vertical scroll container. Messages are aligned based on role:

- **Nexi messages:** Left-aligned with a subtle card background. They render markdown-like formatting (bold, code, lists, math notation). Below each Nexi message, citation badges show which sources were referenced. Two action buttons appear: Save to Notebook and Copy.
- **User messages:** Right-aligned with a primary-colored background and lighter text.

Input area

A textarea at the bottom with auto-height expansion. It supports Enter to send and Shift+Enter for new lines. A microphone button toggles voice input (simulated: shows a pulsing indicator and generates fake transcript text). A send button appears when text is present.

Follow-up chips

After each Nexi response, a row of follow-up action chips appears: Explain Simply, Quiz Me, Extract Key Ideas, Compare Concepts. Each has an associated icon. Clicking a chip sends it as a user message and triggers a simulated Nexi response.

Simulated responses

Nexi responses are not powered by an AI model. They are pre-written responses stored in simulatedResponses in WorkspaceContext.tsx. The follow-up chips map to specific detailed responses. Unknown prompts receive a generic response. All simulated responses include citations and are written in the style of a patient, knowledgeable tutor.

Save to Notebook

Clicking the bookmark icon on a Nexi message saves it as a notebook entry linked to the current course, sourced as Nexi response. A toast confirms the save.

Copy

Clicking the copy icon copies the message content to the clipboard with a toast confirmation.

7.4 Notebook Pane (Right)

The right pane is a lightweight notebook sidebar within the workspace.

Mini mode (56px)

Shows a toggle button and a BookOpen icon with a count badge showing the number of saved notes.

Expanded mode (300px)

Shows a header with note count, a quick capture text input (type and press Enter to add a quick note), and a scrollable list of existing notes. Each note shows title, snippet, up to 2 tags, and date.

This pane shares the same notebookEntries state as the full Notebook page. Notes added here appear on the full page and vice versa.

8. Interaction and State Logic

8.1 Shared State (WorkspaceContext)

All meaningful application state lives in WorkspaceContext, provided by WorkspaceProvider in Layout.tsx. This ensures state is shared across all pages within a session:

- Adding a note in the workspace notebook pane immediately reflects on the full Notebook page.
- Completing a task on the Home page immediately updates the Study Plan page.
- Changing font family in Settings immediately affects all pages.

The context exposes both state and update functions (addNotebookEntry, toggleTask, updateAppSettings, etc.).

8.2 Theme State

Theme is managed separately from WorkspaceContext, in ThemeProvider. It persists to localStorage and applies CSS classes to the HTML root element. The ThemeProvider exposes theme (preference), resolvedTheme (actual), setTheme, and toggleTheme.

8.3 Pane Coordination

The Workspace page manages pane states locally (not in WorkspaceContext). Source mode changes trigger notebook state changes through useEffect hooks. This creates a coordinated expansion behavior: sources expand, notebook shrinks; sources close, notebook restores.

8.4 Toast Notifications

The app uses the sonner library for toast notifications. Toasts appear for: saving notes, copying text, completing tasks, changing settings, language changes, and various confirmations. They auto-dismiss and appear in the bottom-right.

8.5 What is Static vs Dynamic

Dynamic (changes in response to user action): task list, notebook entries, reflections, chat messages, user profile, app settings (font, theme, compact mode, language), pane expand/collapse states.

Static (hardcoded): course catalog, source documents, weekly activity data, progress metrics, session list, connector list, Nexi simulated responses.

9. Motion and Polish

9.1 Implemented Motion

- Fade-in animations on page load (animate-fade-in with staggered delays via animation-delay).
- Slide-up animations for expanding panels (animate-slide-up on task form, sync panel).
- Scroll reveal: The useScrollReveal hook uses IntersectionObserver to trigger fade-in when elements enter the viewport (used on Progress page).
- Spring easing (ease-spring custom timing function) on pane width transitions, icon rotations, and interactive feedback.
- Active press states: scale 0.95–0.97 on all interactive elements (buttons, cards, toggles).
- Hover states: subtle background change, shadow lift on cards (card-interactive class), color transitions on icons and text.
- Progress bar animation: bars animate from 0% to target width when visible, with configurable delays.
- Checkbox pop animation (animate-check-pop) when completing tasks.
- Toggle switches use Apple-style sliding thumb animation.
- The Plus icon on Add Task rotates 45° to become an X when the form is open.

9.2 What Feels Polished

- Settings page card pickers with hover scale, accent borders, and checkmark badges.
- Sidebar collapse/expand with icon-rail mode.
- Task management interactions (add, check, delete with hover reveal).
- Nexi chat with citations and follow-up chips.
- Toast feedback throughout the app.

9.3 What Still Feels Like Prototype

- Progress page data is all fake and the chart is basic divs.
- Source viewer has placeholder content for videos and slides.
- Nexi responses are pre-written, not AI-generated.
- No loading states or skeleton screens (all data is instant from memory).
- Empty states exist but some are minimal.

10. Today's Refinements and Decisions

This section documents all changes discussed, planned, and implemented during today's working session.

10.1 Sidebar Navigation Reorder

The six sidebar navigation items were reordered for better logical flow:

Before: Home, Library, Notebook, Reflections, Progress, Study Plan

After: Home, Library, Study Plan, Notebook, Progress, Reflections

Rationale: Library (course directory) naturally comes after Home. Study Plan (what to do next) comes before Notebook (capture knowledge). Progress (review) comes before Reflections (introspect). This follows a natural learning workflow: find courses, plan study, take notes, check progress, reflect.

10.2 Settings Page Redesign

The General tab of Settings was completely redesigned from simple toggle rows to visual card-based pickers:

- Color mode cards with mini-preview illustrations (warm paper for light, split for auto, charcoal for dark).
- Font style Aa cards that render in their actual font family.
- Both card sections use accent-colored borders and checkmark badges for selection state.
- Hover effects: slight scale up. Press effects: scale 0.97.
- Additional sections added: Voice & Input, Reading & Sources.

10.3 Auto Theme Mode

ThemeProvider was extended from 2 modes (light/dark) to 3 (light/dark/auto):

- Auto resolves to the OS preference via prefers-color-scheme media query.
- A real-time listener updates the theme if the OS preference changes while on auto.
- The toggle button in the sidebar cycles light to dark. In auto mode, it resolves to the opposite of current.
- LocalStorage key: nexus-theme (stores light, dark, or auto).

10.4 Font Family System

A complete font family preference system was added:

- Four options: Default (Source Serif 4 headings + Inter body), Sans (Inter everywhere), Serif (Source Serif 4 everywhere), Dyslexic (OpenDyslexic everywhere).
- fontFamily added to AppSettings in WorkspaceContext.

- Layout.tsx applies font-family-sans, font-family-serif, or font-family-dyslexic CSS class to root.
- index.css defines these classes with appropriate font stacks.
- OpenDyslexic and Source Serif 4 loaded via Google Fonts @import.

10.5 Sidebar Language Submenu

The Language menu item in SidebarUserMenu was converted from a simple menu item to a dropdown submenu:

- Lists 10 languages: English, Español, Français, Deutsch, Português, Japanese, Chinese, Korean, Arabic, Hindi.
- Selected language shows an accent-colored checkmark.
- Selection updates appSettings.language via WorkspaceContext and shows a toast confirmation.
- This preference is intentionally separate from the Nexi response language in Settings. The sidebar controls UI language preference; the Nexi tab controls AI response language.
- Full UI translation (i18n) was deliberately deferred as it requires translating every string in the app.

10.6 Nexi Settings Tab

A dedicated Nexi tab was added to Settings with controls for: response style (concise/detailed/socratic), citation visibility, follow-up suggestions, conversation memory, code explanation depth (beginner/intermediate/advanced), and preferred response language.

11. Current Gaps and Next Priorities

11.1 Status Overview

Feature	Status	Notes
Home page	Complete	Greeting, tasks, courses, quick actions
Library page	Complete	Search, filters, course cards
Study Plan page	Complete	Task CRUD, date/time, priority, sync UI
Notebook page	Complete	Editor, search, sort, view modes
Progress page	Needs work	All data hardcoded, basic chart
Reflections page	Complete	Mood, voice, date grouping, prompts
Settings page	Complete	5 tabs, visual pickers (redesigned today)
Profile page	Complete	View/edit, stats
Workspace layout	Complete	3-pane with coordinated expand/collapse
Sources pane	Mostly complete	Viewer has placeholders for video/slides
Nexi pane	Prototype	Simulated responses only, no real AI
Notebook pane	Complete	Quick capture, note list, shared state
Theme system	Complete	Light/Dark/Auto with persistence
Font system	Complete	4 families, 3 sizes, global application
Language selection	UI only	No actual i18n translation
Data persistence	Missing	All state resets on page refresh
Authentication	Missing	No login/signup, hardcoded user
Backend/API	Missing	No server, no database
Real AI integration	Missing	Nexi uses pre-written responses
Connector integrations	UI only	Buttons exist, no real connections

11.2 Priority Next Steps

Based on the current state, the recommended next priorities are:

- **1. Data persistence:** Connect to a backend (Supabase via Lovable Cloud) to persist notes, reflections, tasks, and settings across sessions.
- **2. Authentication:** Add login/signup so users have actual accounts. This unlocks multi-device access and real profile management.

- **3. AI integration:** Replace simulated Nexi responses with a real language model, grounded in uploaded course materials via RAG (retrieval-augmented generation).
- **4. Source uploads:** Allow users to actually upload PDFs, documents, and links instead of relying on hardcoded source data.
- **5. Progress page redesign:** Replace hardcoded metrics with real data derived from actual usage (notes created, time spent, courses visited).
- **6. Mobile responsiveness:** The current layout is desktop-focused. The workspace 3-pane layout especially needs a mobile strategy.

12. Final Summary

Nexus² is a well-structured frontend prototype of a privacy-first learning companion. The UI is thoughtfully designed with warm, scholarly aesthetics and careful attention to interaction quality. The 3-pane workspace, multi-page note management, task tracking, reflections journal, and configurable settings create a comprehensive learning environment.

Today's session focused on visual refinement: redesigning the Settings page with card-based pickers, adding Auto theme support, implementing a font family selection system, creating a sidebar language submenu, and reordering navigation for better logical flow.

The primary gap is the absence of a backend. All data lives in React state and disappears on refresh. The AI tutor uses pre-written responses. Once persistence, authentication, and real AI integration are added, the product will transition from an interactive prototype to a functional learning tool.

The design foundation is strong. The warm color palette, serif+sans typography, privacy-first messaging, and no-gamification stance create a distinctive identity that sets it apart from typical EdTech products. The next phase should focus on infrastructure (backend, auth, AI) while preserving this design quality.